

**Syllabus
Academic Year 2024-2025**

1. General Information	
Course title	Big Data and Distributed Algorithms
Degree cycle (level)/major	Bachelor degree / 6B06103 Big Data Analysis
Year, term	Year 3, Term 2
Number of credits	5
Language of delivery:	English
Prerequisites	“Big Data and Distributed Algorithms” is based on knowledge obtained by the students on Information and Communication Technologies, Database Management Systems, Computer Organisation and Architecture, Machine Learning, Cloud Computing, and Operating Systems and Computer Networks courses.
Postrequisites	Knowledge obtained during the course will be used for big data analysis and performing research on data analytics.
Lecturer(s)	Yermek Alimzhanov, senior lecturer, ermek.alimzhanov@astanait.edu.kz Astana IT University, Expo, C1 block, 3d floor, office C1.2.327.
2. Goals, Objectives and Learning Outcomes of the Course	
1. Course Description	This course introduces you to the core concepts of big data processing and analytics. The course helps learners develop skills with cloud services that are critical for conducting analysis of big data problems. This course is intended for students who seek expertise on the tasks, tools, and strategies that are used to collect, store, prepare, analyze, and visualize data for use in analytics and ML applications. This course is most aligned to a data engineer role but would also be appropriate for data analysts; data scientists; extract, transform, and load (ETL) developers; or ML practitioners who want to understand how the data that they use in their analyses and predictions is prepared for analysis. The course will also familiarize you with some distributed algorithms and their application areas. You will learn about important big data storage and processing frameworks like Apache Hadoop, Apache Spark, and other software packages of Hadoop ecosystem.
2. Course Goal(s)	This course is designed to give students a detailed overview of methods and tools for Big Data collection, storage, processing, and analysis. Throughout the course, students will explore use cases from real-world applications, which will enable them to make informed decisions while building data pipelines for their applications. The goal of the course is to introduce the students to important basic concepts of Big Data analytics and provide practice to obtain a better understanding of the problems. This creates a basis for the students to learn more advanced

	concepts and other technologies in more details in preparation either for a job in industry or for the following courses at the graduate level.
3. Course Objectives:	You will learn: 1. About Big Data and its elements (the five Vs of data). 2. How to design modern data pipeline architecture. 3. On securing and scaling the data pipeline. 4. How to perform ETL by batch or by stream data. 5. Storing and analyzing data using Apache Hadoop framework. 6. Big data processing and analytics with Apache Spark. 7. The machine learning (ML) concepts and lifecycle. 8. Data visualization tools and services. 9. What are distributed systems and algorithms? 10. Preliminary examples of Distributed algorithms and their applications.
4. Skills & Competences	This course requires a strong foundation in IT concepts and skills. To ensure success in this course, students should have the following: - Experience with cloud services (AWS, GCP, etc.). - Worked with databases and Structured Query Language (SQL). - Operating systems and command line interface (Linux, bash, etc.). - Introduced to general networking concepts. - Understanding of decision-making knowledge in math, probability, and statistics.
5. Course Learning Outcomes:	By the end of this course students will be able to: • Summarize the role and value of data science in a data-driven organization. • Recognize how the elements of data influence decisions about the infrastructure of a data pipeline. • Illustrate a data pipeline by using cloud services to meet a generalized use case. • Identify scaling considerations and best practices for building pipelines that handle large-scale datasets. • Design and build a data collection process while considering constraints such as scalability, cost, fault tolerance, and latency. • Implement the steps to process structured, semistructured, and unstructured data formats in a data pipeline. • Explain the concept of MapReduce and how Apache Hadoop is used in big data analytics. • Analyze data by using open-source tools that are appropriate to a given use case.
6. Methods of Assessment	The expected learning outcomes for the course will be assessed through graded and ungraded activities. The graded activities include exam and – Midterm and Endterm quizzes on lectures and labs.

	– Lab assignments in the AWS Academy Sandbox environments. – Self-study on AWS Academy Data Engineering (ADE) and Knowledge Checks.
7. Reading List	Assigned reading materials and presentations should be read prior to class. Class lectures and discussions will proceed with supplemental and advanced topics, which could be difficult to understand unless students have read the assigned material. Readings are listed in the schedule section. All necessary updates and / or changes to the course will be reflected in the Learning Management System (moodle.astanait.edu.kz). <u>Basic Literature:</u> 1. “EMC Education Services, (2015) "Data Science and Big Data Analytics: Discovering, Analyzing, Visualizing and Presenting Data," Wiley; 1st edition. 432 pages. 2. Jules Damji, et al. (2020) "Learning Spark: Lightning-Fast Data Analytics," O'Reilly Media, 2nd edition. 397 pages. 3. Jeffrey Dean, Sanjay Ghemawat (2004) MapReduce: Simplified Data Processing on Large Clusters, Google. 4. Sakti Mishra, (2022) "Simplify Big Data Analytics with Amazon EMR: A beginner's guide," Packt Publishing, 430 pages. 5. Ajay D. Kshemkalyani, Mukesh Singhal. (2008) "Distributed Computing: Principles, Algorithms, and Systems." Cambridge University Press, 1st edition. 754 pages. <u>Supplementary literature:</u> 1. Jiawei Han, Micheline Kamber, Jian Pei. "Data mining: concepts and techniques", Morgan Kaufmann Publishers (3rd ed.), 703 pages. 2. Gerard Tel, Introduction to Distributed Algorithms, Cambridge University Press; 2nd edition (612 pages, 2000)
1. Resources	2. https://www.awsacademy.com/ (ADE) 3. https://www.hadoopinrealworld.com/starterkit/ 4. https://community.cloud.databricks.com/
5. Course policy	Course and University policies include: Attendance: Attendance is not allocated any grading points in the marking scheme, but is compulsory to pass the course. Normally students are required to achieve course attendance of minimum 70% to get admitted to the examination rubric. In case a student misses 30% or more class sessions without a valid excuse the instructor has the right to mark him as “not graded”. In such case a student is not admitted to the exam and automatically fails the course. In cases, when a student misses class session due to valid reasons (is excused by the instructor or the dean’s office) he or she has to confirm the absence reason using a valid document in accordance with the academic policy of AITU.

It should be NOTED that in cases when a student is excused for 30% of the scheduled class sessions or more he or she has to study material provided under the course on their own. Course instructor might provide additional opportunities to submit missed graded pieces of work during office hours or conduct alternative assessment exercises using method of his or her choosing.

Preparation for Class: Class participation is a very important part of the learning process in this course. Although not explicitly grade, students will be evaluated on the QUALITY of their contributions and insights. Quality comments possess one or more of the following properties:

- Offers a different and unique, but relevant, perspective;
- Contributes to moving the discussion and analysis forward;
- Builds on other comments.

Class work: The duration of each lecture and practical lesson is 50 minutes for offline class, and 40 minutes for online class. Students are expected to complete all readings and assignments ahead of time, attend class regularly and participate in class discussions. In case of systemic student's misconduct, the student can be dispensed from the classes.

Being late on class: When students come to class late, it can disrupt the flow of a lecture or discussion, distract other students, impede learning, and generally erode class morale. Moreover, if left unchecked, lateness can become chronic and spread throughout the class. Therefore, the being late to the class is not welcome and can have restriction activities by the course instructor.

Attestation I and II: Students, who score less than 25% for Attestation period I or Attestation period II (RK1/RK2) automatically fail the course.

Home work / Assignments: The assignments are designed to acquaint students with the theoretical knowledge and practical skills required for the course. The textbook readings will be supplemented with materials collected from recent professional articles and journals. In case of using someone's work (papers, articles, any publications), all works must be properly cited. Failure to cite work will be resulted as a cheating from the students and may be a subject of additional disciplinary measures.

Late submissions: Most assignments will be discussed in class on the due date. It is expected that all work will be submitted on time. All gradings are based using a percentage grading scale.

In the case of some extraordinary event, students should notify the course instructor and request an extension of the deadline for submission. If approved, a new date will be given to the student depending upon the circumstances by the instructor.

Final exam* / Final project:

	The final exam for the course will be taken in the form of a computer test organized with the Learning Management System (moodle.astanait.edu.kz). Laptops and mobile devices can only be used for classroom purposes when directed by the course instructor. Misuse of laptops or handheld devices will be considered a breach of discipline and appropriate action will be initiated by the instructor. Online lessons can be used in case if there won't be a chance to make offline traditional lessons. It must not discourage the interest and enthusiasm of students. The main software to run the online lessons is Microsoft Teams for video calls and live webinars, and Moodle (moodle.astanait.edu.kz) as a Learning Management System. Also, some alternatives such as Telegram, Zoom, or other messenger may be involved as an additional workaround. Cheating and plagiarism are defined in the Academic conduct policies of the university and include: 1. Submitting work that is not your own papers, assignments, or exams; 2. Copying ideas, words, or graphics from a published or unpublished source without appropriate citation; 3. Submitting or using falsified data; 4. Submitting the same work for credit in two courses without prior consent of both instructors. Any student who is found cheating or plagiarizing on any work for this course will receive 0 (zero) for that work and further actions will also be taken regarding academic conduct policies of the university. Academic Conduct Policies of the university: The full texts of all the academic conduct code will be posted to the students using Learning Management System (moodle.astanait.edu.kz). Contacting the Course instructor: The easiest and most reliable way to get in touch with the course instructor is by email or MS Teams. Students must feel free to send email if you have a question related to the course. Instructor responds as soon as they can but not always instantaneously. Besides that, students are also welcomed to arrange a one-to-one meeting with the instructor by their office during office hours to discuss the class using both offline and online.
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** Final exam can be given by instructor in the any other forms such as Written Exam, Final Quiz, Oral Tasks and Exams, etc.*

3. Course Content

#	Abbreviation	Meaning
1	ISIS	Instructor-supervised independent work
2	SIS	Students' independent work
3	IP	Individual project
4	PA	Practical assignment
5	LW	Laboratory work
6	MCQ	Multiple choice quiz

3.1 Lecture, Practical/Seminar/Laboratory Session Plans

Week No	Course Topic	Lectures (H/W)	Practice sessions (H/W)	Lab. sessions (H/W)	TSIS (H/W)	SIS (H/W)
1	Topic 1. Introduction to Big Data. Data-Driven Organizations.	2	3		1	9
2	Topic 2. Design Principles and Patterns for Data Pipelines.	2	3		1	9
3	Topic 3. Securing and Scaling the Data Pipeline.	2	3		1	9
4	Topic 4. Ingesting and Preparing Data. ETL and ELT comparison. Ingesting Data by Batch or by Stream.	2	3		1	9
5	Topic 5. Storing and Organizing Data. Data lake and data warehouse storages.	2	3		1	9
6	Topics 6. Apache Hadoop and its ecosystem. Processing Big Data with Hadoop MapReduce.	2	3		1	9
7	Topics 7. Introduction to Apache Pig and Apache Hive.	2	3		1	9
8	Topic 8. Processing Data for ML. Analyzing and Visualizing Data.	2	3		1	9
9	Topic 9. Big Data analysis with Apache Spark framework.	2	3		1	9
10	Topic 10. Introduction to Distributed Systems and Algorithms.	2	3		1	9
	Total hours: 150	20	30		10	90

3.2 List of Assignments for Student Independent Study

№	Assignments (topics) for Independent study	Hours	Recommended literature and other sources (links)	Form of submission
1	2	3	4	5
1	Lab 1: Accessing and Analyzing Data by Using Amazon S3 in Module 2.	9	[1] Chapter I. ADE, video lectures of Modules 2-3	Labs in the AWS Academy Data Engineering

				(ADE) course. Lab Assignment 1
2	Lab 2: Querying Data by Using Athena in Module 4	9	[1] Chapter II. ADE, video lectures of Modules 4-5	Lab Assignment 2
3	Lab 3: Performing ETL on a Dataset by Using AWS Glue in Module 7	9	[1] Chapter III. ADE, video lectures of Modules 6	Lab Assignment 3
4	Lab 4: Storing and Analyzing Data by Using Amazon Redshift in Module 8	9	[1] Chapter IV. ADE, video lectures of Module 7	Lab Assignment 4
5	Lab 5: Processing Logs by Using Amazon EMR in Module 9	9	[1] Chapter V. ADE, video lectures of Module 8	Lab Assignment 5
6	Lab 6: Updating Dynamic Data in Place in Module 9	9	[1] Chapter VI ADE, video lectures of Module 9.	Lab Assignment 6
7	Lab 7: Analyzing and Visualizing Streaming Data with Kinesis Data Firehose, OpenSearch Service, and OpenSearch Dashboards in Module 11	9	[1] Chapter X [3] MapReduce ADE, video lectures of Module 10. Hadoop in Real World (see resources)	Lab Assignment 7
8	Lab 8: Building and Orchestrating ETL Pipelines by Using Athena and Step Functions in Module 12	9	[1] Chapter VII ADE, video lectures of Module 11	Lab Assignment 8
9	Capstone Project in the ADE course	9	[2] Learning Spark, Pages 1-17, 43-112. ADE, video lectures of Module 12	ADE, Capstone Project.
10	Course Assessment quiz in the ADE course.	9	[1] Chapter XII	ADE, Course Assessment.

4. Student Performance Evaluation System for the Course

Period	Assignments	Number of points	Total
1 st attestation	Midterm Quiz Lab assignments Students' independent work (Knowledge checks in Data Engineering course)	20 50 30	100
2 nd attestation	End term Quiz Lab assignments Capstone Project Students' independent work (Knowledge checks in Data Engineering course)	20 40 20 20	100

Final exam*	<i>Final Quiz</i>		100
Total	0,3 * 1st Att + 0,3 * 2nd Att + 0,4*Final		100

****** *The number of assignments can be different. It depends from the course program and designed by course syllabus. But the total points of the assignment are 60 in each control period.*

***** *Final exam can be given by instructor in the any other forms such as Written Exam, Final Quiz, Oral Tasks and Exams, etc.*

Achievement level as per course curriculum shall be assessed according to the evaluation chart adopted by the academic credit system.

Letter Grade	Numerical equivalent	Percentage	Grade according to the traditional system
A	4,0	95-100	Excellent
A-	3,67	90-94	
B+	3,33	85-89	Good
B	3,0	80-84	
B-	2,67	75-79	
C+	2,33	70-74	
C	2,0	65-69	Satisfactory
C-	1,67	60-64	
D+	1,33	55-59	
D	1,0	50-54	
FX	0	25-49	Fail
F	0	0-24	

Based on the specific grade for each assignment, and the final grade, following criteria must be satisfied:

Grade	Criteria to be satisfied
90-100	- Work would be worthy of further dissemination under appropriate conditions - Mastery of advanced methods and techniques at a level beyond that explicitly taught - Ability to synthesize and employ in an original way idea from across the subject - Outstanding command of critical analysis and judgment
80-89	- Excellent range and depth of attainment of intended outcomes - Mastery of a wide range of methods and techniques - Evidence of study and originality of what has been taught - Able to display a command of critical analysis and judgement
70-79	- Attained all the intended learning outcomes for a unit - Able to use well a range of methods and techniques to come to conclusions - Able to employ critical analysis and judgement

60-69	- Some limitations in attainment of learning objectives, but has managed to grasp most of them - Able to use most of the methods and techniques taught - Evidence of study and comprehension of what has been taught but grasp insecure - Some grasp of the issues and concepts underlying the techniques and material taught, but weak and incomplete
50-59	- Attainment of only a minority of the learning outcomes - Able to demonstrate a clear but limited use of some of the basic methods and techniques taught - Weak and incomplete grasp of what has been taught - Deficient understanding of the issues and concepts underlying the techniques and material taught
25-49	- Attainment of nearly all the intended learning outcomes deficient - Lack of ability to use at all or the right methods and techniques taught - Inadequately and incoherently presented - Wholly deficient grasp of what has been taught - Lack of understanding of the issues and concepts underlying the techniques and material taught
0-24	No significant assessable material, absent or assessment missing a must pass component

5. Methodological Guidelines

Assessment is administered continuously throughout the course. The students are rated against their performance in continuous rating administered throughout the semester (60%) and summative rating done during the examination session (40%), total 100%. Continuous rating is students' on-going performance in class and independent work. Class work is assessed for attendance, laboratory works' defense and in- class assessments.

- **ISIS (Instructor Supervised Student Independent Study)** -comprises presentation to be done by students independently and checked by instructor.
- **Mid-term and End-term** is a review of the topics covered and assessment of each student's knowledge. The form of the midterm and end term exams is complex.
- **Final assessment** is a combination of both individual (team) project (report) and oral presentation (slides) to evaluate the students' academic performance and professional skills. At the completion of this course each student has to submit an online project version conforming to the project outline, as well as to prepare and present a slide presentation that follows the presentation outline. Project iterations would be required to submit as well. Students should submit the written Report of Final project and slide-Presentation 3 days before the Final exam day.

6. Lecturer (lecturers) approvals Full name Job title Date Sign

Full name	Job title	Date	Sign
Yermek Alimzhanov	Senior Lecturer	_____ 2024	

Director

Sergaziyev M.Zh.